

HTML



UX

SEO

HTML 5

CMS



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Objective:

After completing this unit, Learner will be able to :

- Develop static website using different HTML Controls.
- Able to Create interactive Form to accept data from user.
- Able to represent data/information in different layouts and formats.

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Introduction to HTML

- HTML (***Hypertext Markup Language***) is used to create web document known as Web Page.
- HTML is ***not a programming language***, it is a Markup Language .
- It is simply a collection of certain keywords called “Tags”.
- HTML is not a case sensitive language means we can write the code of html in upper case or in lower case.
- ***Tim Berners-Lee*** created HTML in late 1991 but didn't release it officially. He published it later in 1995 as HTML 2.0.
- The latest standard of HTML is **HTML 5.0**

HTML Tags

- HTML Tags are signals to a browser that should do something other than just through text upon the screen.
- Tags are basic building block element of HTML.
- All HTML tags begin with an open angle bracket (<) and end with close angle bracket (>).
- HTML tags can be a two types :-
 - Paired Tag
 - Singular Tags

Paired Tag

- Tags which have both the **opening and closing tag** known as Paired tag.
- Paired tag also known as **container tags** because they hold some **text or another HTML tags** in between it.

Example:

- `<body></body>`
- `<p></p>`
- `<b></b>`

Empty Tags

- Tags, which have only opening and no ending, are called empty tags.
- Empty tags also known as singular tags.
- Example:
 - `<hr/>`
 - `<br/>`
 - `<img/>`

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Structure of HTML page

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

Some tags are placed here..

```
  </head>
```

```
  <body>
```

Some text/tags are placed here..

```
  </body>
```

```
</html>
```

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HTML Tag <HTML>

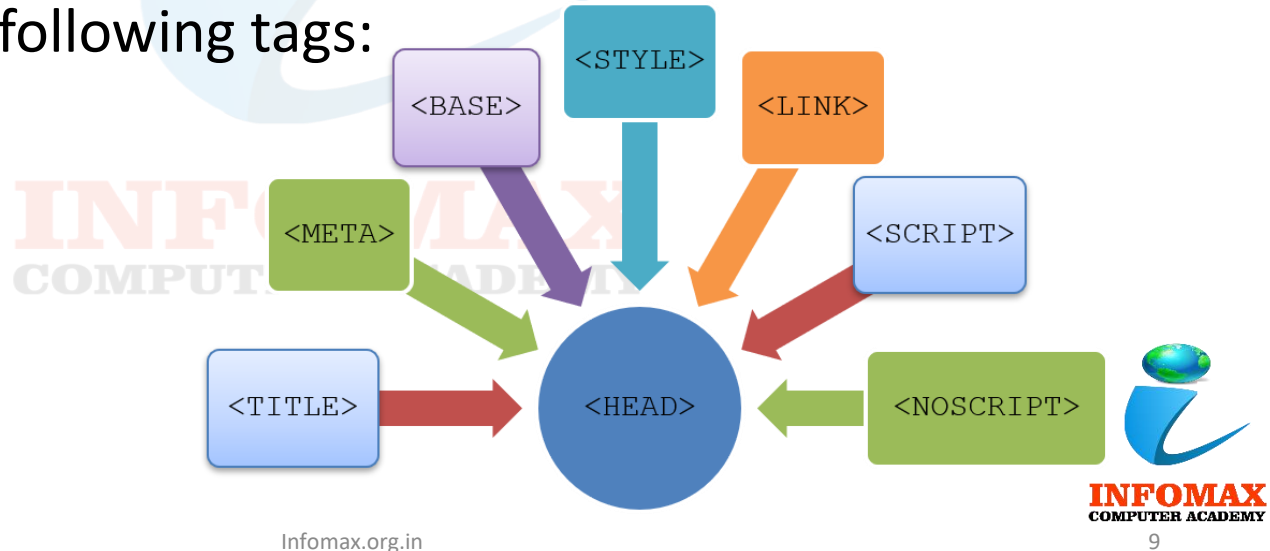
- The <HTML> tag encloses all other HTML tags and associated text within your document.
- It is an optional tag.
- You can create an HTML document that omits these tags, and your browser can still read it and display it. But it is always a good form to include the start and stop tags.

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Head Section and Elements of Head Section

The <HEAD> tag:

- Is contained within the <HTML> and </HTML> tags.
- Is used to describe the header of the HTML document.
- Can include the following tags:



Elements of Head Section

The <TITLE> tag:

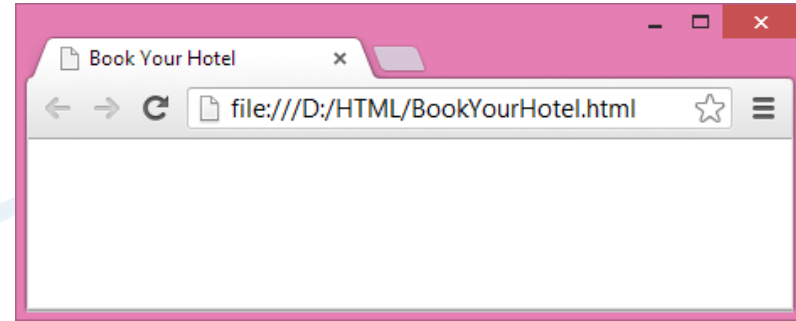
- Defines the title of the document.
- Can occur only once.

For example:

```
<HEAD>
```

```
<TITLE> Book Your Hotel</TITLE>
```

```
</HEAD>
```



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The body section

- Is the central part of the HTML document required to display content on the Web page.
- Begins with the <BODY> tag.
- Contains the actual visible content of an HTML document.

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Attribute of Tag

- Attribute are used to say something about the element that carries them and it always appear on the opening tag of the element that carries them.
- All attributes are made-up two parts a name and a value.
 - The name is the property of the element that you want to set.
 - The value is what you want the value of property to be.

Syntax:

`<tagName attributeName="value">content</tagName>`

Example:

`<p align="center"> Text</p>`

Attributes of Body Tag:

BGCOLOR

- To set the background color page (color name or color code)

TEXT

- To set the text color of web page (color name or color code)

TOPMARGIN

- To set the margin from top of page (px or %)

LEFTMARGIN

- To set the margin from left & right of page (px or %)

BACKGROUND

- To set the background image on web page (url/path)

LINK

- To set the link color of page (color name or color code)

VLINK

- To set the color of visited link of page (color name or color code)

ALINK

- To set the link of active link of web page (color name or color code)

ALIGN

To set the alignment of body content (left, right, center)

Formatting Tags:

Heading Tags: Specifies the main idea of the content. From `<h1></h1>` to `<h6></h6>` tag used to define different level of headings.

- Attribute of heading tag:
 - align: to set the alignment of heading [left,right,center]
 - Ex: `<h1 align="center"> Heading 1</h1>`

Paragraph: `<p></p>` tag is used to define a paragraph.

- Attribute of `<p>` tag:
 - align : to set the alignment of paragraph [Left, Right, center, Justify]
 - Ex: `<a align="center"> Paragraph text</p>`

Formatting Tags (Conti.)

Link Break : `<br/>` tag is used for manual line break. It is an empty (Singular) tag.

Draw a horizontal line: `<hr/>` tag is used to draw a horizontal line in page.

- Attribute of `<hr/>` tag:
 - color: to change the color of line
 - width: to set the width of line (default is 100%)
 - size: to set the thickness of line
 - align: to set the alignment of line i.e. left, right or center (default is center)
 - noshade : to remove the shade of line (make a solid line)

Text Formatting Tags

- **Bold** : `<b></b>` or `<strong></strong>` Bold Text
- **Italic** : `<i></i>` Italic Text
- **Underline** : `<u></u>` Underline text.
- **Strikethrough** : `<strike></strike>` Line through on text
- **Superscript** : `<sup></sup>` Superscript
- **Subscript** : `<sub></sub>` Subscript.
- **Big** : `<big></big>` Big size Text
- **Small** : `<small></small>` small text
- **Mark** : `<mark></mark>` Highlight text.
- **Insert** : `<ins></ins>` insert text just like underline.
- **Pre** : `<pre></pre>` Preformatted text.

Formatting Tags (Conti.)

** Tag** : To set the text color, size and face(font family)

- Attribute of tag:
 - Color : to set the color of text
 - Size : to set the size of text (range 1 to 7 , default is : 3)
 - Face : to set the font of text ex Times New Roman

Ex: *
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Formatting Tags (Conti.)

Scrolling Text/Image : `<marquee></marquee>` tag is used to scroll text text/image on screen like running news on screen.

- Attribute of `<marquee>` tag:
 - **Direction** : indicate the direction of scrolling text (**left**, right, up, down) .
 - **Scrollamount** : to set the speed of scrolling text .(default :6)
 - **Width** : to set the width of scrolling area
 - **Height** : to set the height of scrolling area
 - **Behavior** : indicate the type of scrolling (Slide, **scroll**, alternate)
 - **Bgcolor** : to set the background color of scrolling text

HTML Entities

HTML entities are used to display reserved characters in HTML.

Symbol	Entity	Symbol	Entity
Space	 	← (LEFTWARDS ARROW)	←
< (Less than)	<	↑ (UPWARDS ARROW)	↑
> (Greater than)	>	→ (RIGHTWARDS ARROW)	→
& (ampersand)	&	↓ (DOWNWARDS ARROW)	↓
© (copyright)	©	π (Pi)	π
® registered trademark	®	₹ (Rupee)	₹
½	½	¼	¼
⅔	⅔	⅓	⅓

Comment in HTML

Comments are text notes added to the program to provide explanatory about the source code.

Syntax:

<!-- Your Comment goes here -->

Example:

<!-- About Section Start here -->

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Image in HTML

- Images can be placed in a web page by using tag.
- The 'src' Attribute is used to set the source of image. The url can be a relative or absolute.
- There are many Image formats available today, but the most widely used among them are .jpg, .jpeg, .png and .gif.

Syntax: ``

Example: ``

Image in HTML (Conti.)

- Attribute of Image tag:
 - Src : to set the source of image (URL/path)
 - Height : to set the height of image (Px or %)
 - Width : to set the width of image (Px or %)
 - Align : to set the alignment of image (Left, Right, center)
 - Border: to define the border of image (Integer Number)
 - Hspace: to set the horizontal space image (Px or %)
 - Vspace: to set the vertical space from image (Px or %)
 - Title : to define the title of image
 - Alt : to set the alternate text for image.

Link in HTML

<a> ANCHOR TAG:

- Web pages are linked to one another through Hypertext Links.
- Text or image in the HTML document can be linked to an external document or to a specific place within the same document.
- The text or image that provides such linkage is known as Hypertext or Hotspot.

Syntax: Text/Image

Example: <a href="<http://google.com>" />Open google



Link in HTML (Conti.)

- **Attribute of Anchor tag:**

- href : to set the link path/source (url, path,file name)
- name : to specify the name of link as bookmark
- target : to set the target of link (_blank, frame name)

- **Link from website:**

- `<a href="http://infomaxacademy.com"> Infomax Academy</a>`

- **Link from other webpage:**

- `<a href="about.html"> About us</a>`

- **Link in same document:**

- `<a name="top"></a>`
- `<a href="#top"> Go to Top</a>`

Link with Email

- HTML <a> tag provides a option to specify an email address to send an email.
- To link a text with email, use mailto: email_address along with href attribute.

`<a href="mailto:abc@example.com"> Send Email</a>`

- We can also specify a default email subject and email body along with email address.

`<a href="mailto:abc@example.com?subject=Feedback&body=Message"> Send Mail</a>`

Link with Mobile/Phone

- To link a text with mobile or phone number, use tel: mobile_number along with href attribute.

`<a href="tel:+23832884"> Call Us</a>`

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Image Linking

Syntax:

```
<a href="url/path"></a>
```

Example:

```
<a href="http://infomaxacademy.com">  
    
</a>
```

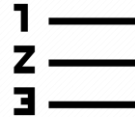
List in HTML:

- HTML Supports several ways of arranging items in lists.
- Type of List:

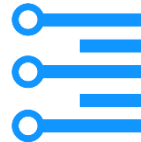
Unordered List



Ordered List



Definition Lists



List in HTML : Unordered List

- Unordered List also called as *bulleted list*, used to present list of items marked with bullets.
- An unordered list starts with in followed by (List Item) tag.

- Syntax:

list Item 1

 List item 2



- Attribute of Tag:

- Type: to set the bullet style.(disk, square, circle) Default – disk

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List in HTML : Ordered List

- Ordered list also called as Numbered list, is used to present a numbered list of item in the order of importance or the item (paragraph) is marked with a number.
- An ordered list must begin with the followed by an list item tag.

- Syntax:

```
<ol>
```

```
    <li>list Item 1</li>
```

```
    <li> List item 2</li>
```

```
</ol>
```



List in HTML : Ordered List (Conti.)

- Attribute of Tag:
 - **Type** : to define the number style (A,a,I,I,1)
 - **Start** : to set the start value of numbering (any interger)
- Attribute of tag:
 - **Value**: to change the track on numbering (any integer)

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List in HTML : Definition Lists

A definition list is a list of terms and corresponding definitions. Definition lists are typically formatted with the term on the left with the definition following on the right or on the next line. The definition text is typically indented with respect to the term.

The opening list tag must be `<DL>`. It is followed by an term names (`<DT>`) and then by definitions (`<DD>`).

Syntax:

```
<DL>
  <dt>Term 1</dt>
  <dd>This is the definition of the first term.</dd>
  <dt>Term 2</dt>
  <dd>This is the definition of the second term.</dd>
</DL>
```


List in HTML : Nested List

- One list might contain one or more ordered/unordered list that is called as Nested lists.



Table in HTML

- Following tag are used to create table:
 - **Table Tag:** The <table> element is used to define a table.
 - **Table Row:** Rows are defined using the <tr> tag (Table Row).
 - **Table Headers:** The <th> tag defines a header cell, typically bold and centered.
 - **Table Data:** Cells in a table are defined using the <td> tag (Table Data).
 - **Table Caption:** The <caption> tag is used to give a title or description to the table. It is placed immediately after the opening <table> tag.
 - **Table Body and Head:**
 - The <thead> tag is used for the table header section.
 - The <tbody> tag is used for the main body of the table.

Example

```
<table border="1">
  <caption>Table 4.5</caption>
  <thead>
    <tr>
      <th>header 1</th>
      <th>header 2</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>row 1, cell 1</td>
      <td>row 1, cell 2</td>
    </tr>
    <tr>
      <td>row 2, cell 1</td>
      <td>row 2, cell 2</td>
    </tr>
  </tbody>
</table>
```

Table in HTML (Conti)

- Attribute of <table> tag:
 - **Height** : to set the height of table
 - **Width** : to set the width of table
 - **Bgcolor** : to set the background color table
 - **Bordercolor** : to set the border color of table
 - **Align** : to set the alignment of table
 - **Cellpadding** : to set the space between text and table border
 - **Cellspacing** : to set the space between two cells
- Attribute of <tr> tag:
 - **Bgcolor** : to set the background color of specified row
 - **Height** : to set the height of specified row

Table in HTML (Conti)

- Attribute of <td> or <th> tag:
 - **Width** : to set the width of specified column
 - **Height**: to set the height of row
 - **Bgcolor**: to set the background color of specified cell
 - **Align** : to set the horizontal alignment of cell text i.e. center, left or right
 - **Valign** : to set the vertical alignment of cell text i.e. top, bottom or center
 - **Rowspan**: to span (merge) the number of rows into one.
 - **Colspan**: to span(merge) the number of column into one.

Form

- HTML forms are used to pass data to a server. A form can contain input elements like text fields, checkboxes, radio-buttons, submit buttons and more.
- The <form> tag is used to create an HTML form:
 <form>
 Form elements (ex. Textbox, radio button, checkbox etc)
 </form>
- Attribute of <Form> tag:
 - **Name** : to specify the name of form
 - **Action**: to set the target location of form when form send
 - **Method** : specify the method for sending data (GET/POST)
 - **Autocomplete** : To enable and disable the autocomplete the from

Form Elements

- **Label :**

- <label> Student Name</label>

- **Text Box :** (To accept single line text)

- <input type="text" name="user_name"/>

- **Password :** (To Accept Secure data like Password/Pin)

- <input type="password" name="user_pass"/>

- **Number Box :** (to accept Numeric Values)

- <input type="number" min="0" max="100" step="10"/>

- **Date :** (To accept Date from Calendar)

- <input type="date" name="dob"/>

Form Elements (Conti)

- **Email** : (to Accept email in valid format)
 - `<input type="email" name="user_email"/>`
- **Search** : (display cross button inside text box which type some text on box)
 - `<input type="search"/>`
- **Month** : to accept only month and year
 - `<input type="month"/>`
- **Time** : to accept Time (Hour minute)
 - `<input type="time"/>`
- **Week** : to accept week and year
 - `<input type="week"/>`

Form Elements (Conti)

- **Tel:** it show number pad when using mobile phone for input value
 - `<input type="tel"/>`
- **url :** to accept a url , it validate the url for valid protocol [http, https].
 - `<input type="url"/>`
- **Color :** to accept color from colorpicker
 - `<input type="color"/>`
- **Range :** to create a range bar
 - `<input type="range" min="1" max="1000" step="10"/>`

Form Elements (Conti)

- **Radio Button:** (To provide Multiple option and single selection)
 - `<input type="radio" name="gender"/>`
 - *Note: to form a group of radio button the name of all radio button must be same.*
- **Checkbox :** to provide multiple option and multiple selection.
 - `<input type="checkbox"/>`
- **Textarea :** to accept multiline text
 - `<textarea rows="5" cols="30"></textarea>`
- **File :** to create file browse button
 - `<input type="file"/>`

Form Element (Conti..)

- **Select Box:** to create a dropdown select box:

```
<select>
```

```
    <option> Item 1</option>
```

```
    <option> Item 2</option>
```

```
    <option> Item n</option>
```

```
</select>
```

- To enable the multiple Selection use the 'multiple' attribute in <select> tag:

```
<select multiple>
```

```
....
```

```
</select>
```

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Form Element (Conti..)

- Optiongroup: used to group related options within a <select> dropdown.

<select>

<optgroup label="Group 1">

<option> Item 1</option>

<option> Item 2</option>

<optgroup label="Group 2">

<option> Item 1</option>

<option> Item 2</option>

</select>

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Form Element (Conti..)

- **Data list:** data list use to filleter text from list of pre defined items:

```
<input list="courses" type="text">
```

```
<datalist id="courses">
```

```
  <option value="CCA">
```

```
  <option value="DCA">
```

```
  <option value="CCC">
```

```
  <option value="Tally">
```

```
  <option value="O level">
```

```
</datalist>
```

Form Element (Conti..)

- **Submit Button:** to submit the form
 - `<input type="submit" value="Save"/>`
- **Reset Button:** to create reset(Clear) Button
 - `<input type="reset" value="Clear"/>`
- **Button :** to create custom button to perform custom operation.
 - `<input type="button" value="Calculate"/>`
- **Image Button:** to Create image button.
 - `<button type="button"></button>`

Common Attributes of <input>

- type Defines input type (text, password, email, file, etc.)
- name Name for form data submission
- value Default value of input
- id Unique identifier
- placeholder Hint text inside input
- required Makes field mandatory
- readonly Field not editable but sent
- disabled Field not editable & not sent

Common Attributes of <input>

- maxlength Max number of characters
- min / max Min/Max value (for number/date)
- step Increment step for number/date
- size Visible width in characters
- autofocus Focus automatically on load
- checked Preselect checkbox/radio
- accept File types allowed for upload

Field set and Legend

- <fieldset>
 - The <fieldset> tag is used to group related elements in a form.
 - The <fieldset> tag draws a box around the related elements.
- <legend>
 - The <legend> tag defines a caption for the <fieldset> element.
- Example:

```
<fieldset>  
    <legend> Basic Details</legend>  
    Form Elements  
</fieldset>
```

Frameset

- HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document.

```
<frameset rows="" >  
    <frame src="url"/>  
    <frame src="url"/>  
</frameset>
```

```
<frameset cols="" >  
    <frame src="url"/>  
    <frame src="url"/>  
</frameset>
```

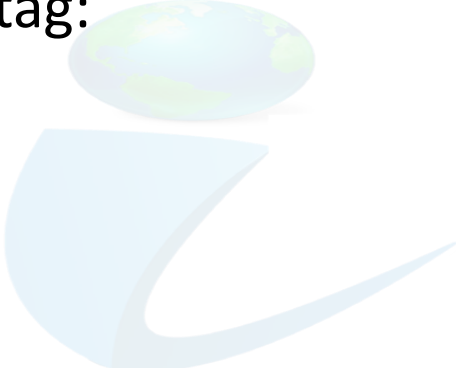
Attribute of <frameset>

- rows
- cols
- border
- bordercolor

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Frameset (Conti.)

- Attribute of <frame> tag:
 - Name
 - Src
 - Border
 - Noresize
 - scrolling



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Iframe

- Iframe known as inline Frame
- An inline frame is used to embed another document within the current HTML document.
- The Document may contain text, video, map etc.
- `<iframe src="url" height="" width="" ></iframe>`
- Others Attribute of `<iframe>`
 - frameborder (yes, no)
 - Scrolling (yes, No , Auto)
 - name

Audio in HTML

- The HTML <audio> element is used to play an audio file on a web page.

<audio controls>

<source src="horse.ogg" type="audio/ogg">

*<source src="horse.mp3"
type="audio/mpeg">*

</audio>

Video in HTML

- The HTML <video> element is used to show a video on a web page.

```
<video width="320" height="240" controls>  
  <source src="movie.mp4" type="video/mp4">  
  <source src="movie.ogg" type="video/ogg">  
</video>
```

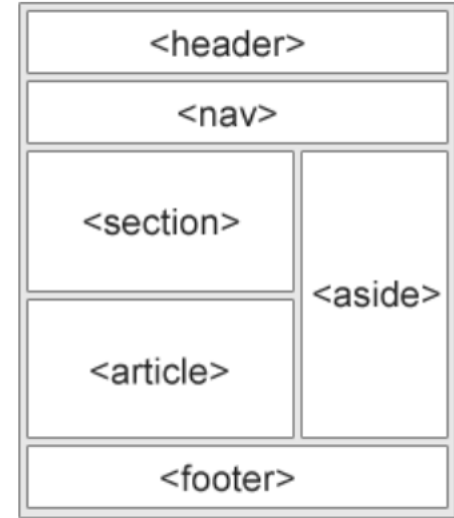
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HTML 5 Tags

- HTML5 introduced several new elements and attributes, enhancing the capabilities of web development.
- Sectioning Elements
- Multimedia Elements
- Interactive Elements
- Form Elements
- Semantic Elements
- Document Metadata
- Canvas
- SVG and MathML

HTML 5: Sectioning Elements:

- <header>
- <footer>
- <nav>
- <article>
- <section>
- <aside>



HTML 5 : Multimedia Elements:

- <audio>
- <video>
- <source>
- <track>

- <canvas>
- <svg>



HTML 5 : Interactive Elements:

- <details>
- <summary>
- <mark>
- <progress>
- <meter>



HTML 5 : Form Elements:

- <datalist>
- <output>
- <input> (range,color,date,etc)
- <textarea>
- <button>
- <select>
- <select>
- <optgroup>
- <option>
- <fieldset>
- <legend>
- <label>

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HTML 5 : Semantic Elements:

- <main>
 - <figure>
 - <figcaption>
 - <mark>
 - <progress>
 - <output>
 - <time>
- <ruby>
 - <rt>
 - <rp>

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HTML 5 Semantic element for Text Formatting

- `<abbr>`: Represents an abbreviation or acronym.

`<abbr title="World Health Organization">WHO</abbr>`

- `<em>`: Represents text with emphasis.

`This is <em>very</em> important.`

- `<cite>`: Represents the title of a creative work (e.g., book, movie, song) or the name of its author.

`<cite>The Great Gatsby</cite> by F. Scott Fitzgerald`

Conti.

- `<pre>`: Represents preformatted text, preserving whitespace and line breaks. Ex.

```
<pre>
```

```
Infomax Computer Academy
```

```
  G. R. Complex Preetam Nagar
```

```
Dhoomanganj, Uttar Pradesh
```

```
  India
```

```
</pre>
```

- `<q>`: Represents inline quotations, typically enclosed within quotation marks.

She said, `<q>I'll be there soon</q>`.

Conti.

- `<blockquote>`: Represents a block of quoted text, often used for longer quotations.

`<blockquote>"The only way to do great work is to love what you do." - Steve Jobs </blockquote>`

- `<dfn>`: Represents a definition term, typically rendered in italics by default.

`<dfn>HTML</dfn>` stands for HyperText Markup Language.

- `<time>`: Represents a specific time or date, optionally with a machine-readable format.

Join us on `<time datetime="2024-03-19">March 19th</time>` for the event.

Conti.

- `<var>`: Represents a variable or placeholder, typically rendered in italics by default.

The formula is `<var>y = mx + b</var>`.

- `<kbd>`: Represents keyboard input, typically displayed in a monospaced font.

To save, press `<kbd>Ctrl + S</kbd>`.

- `<samp>`: Represents sample output from a computer program.

The output of the command is `<samp>hello world</samp>`

- `<code>`: Represents a fragment of computer code, typically displayed in a monospaced font.
Use the `<code>print()</code>` function to display output.
- `<span>`: Provides a generic container for inline elements.
- `<div>`: Provides a generic container for block-level elements.

Thank You

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